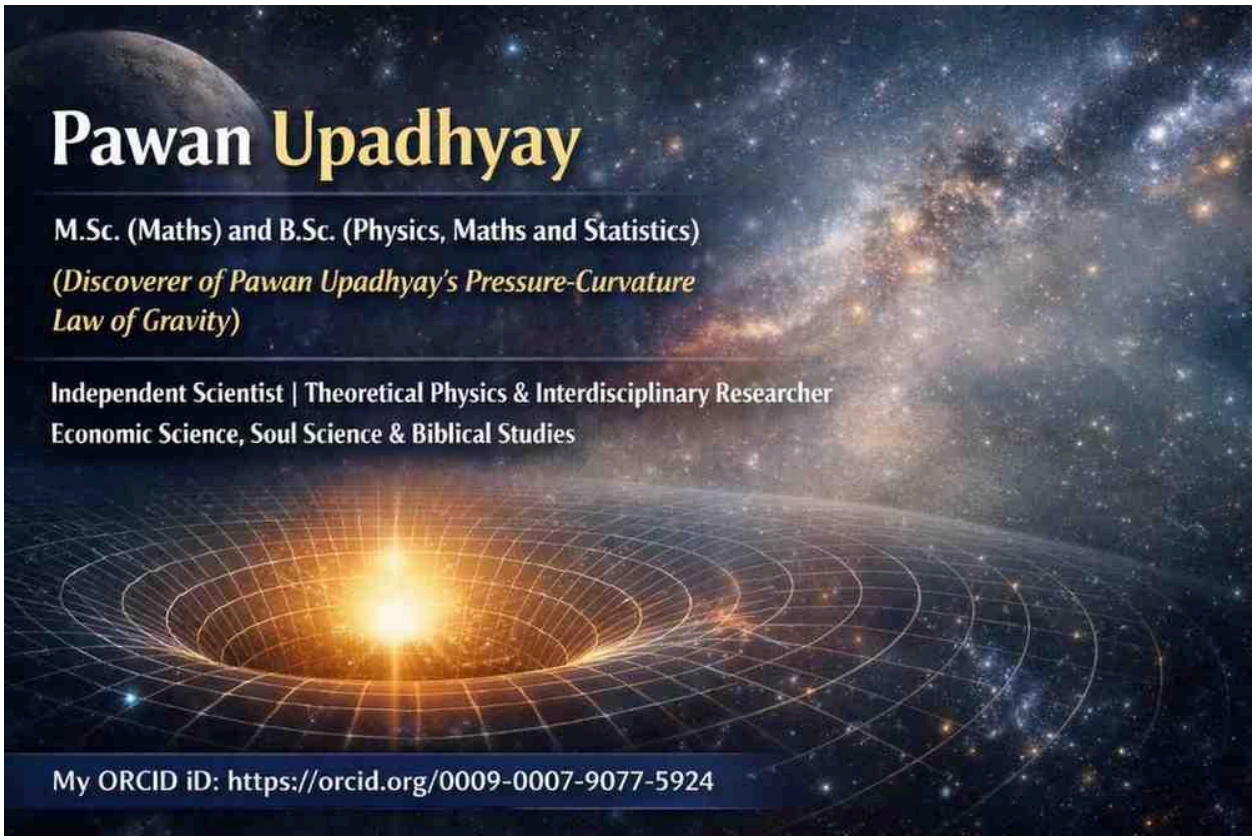
Spacetime Metric Evolution in a Cosmological Fluid with a Variable Equation of State Parameter ω

Author: Pawan Upadhyay
Affiliation: Independent Researcher and Independent Scientist
Email:pawanupadhyay28@hotmail.com
2nd Email: pawankanyakubj@gmail.com
ORCID iD: https://orcid.org/0009-0007-9077-5924



Abstract

The large-scale structure and dynamics of the universe can be modeled by a relativistic perfect fluid characterized by an energy density E_d and pressure P_g. These quantities are related through the equation of state P_g =ωE_d, where ω is known as the equation-of-state parameter. In this study we investigate how the parameter ω affects the spacetime geometry and expansion dynamics of the universe within the framework of general relativity. Starting from Einstein's field equations and assuming a homogeneous and isotropic universe, we derive the Friedmann equations for a cosmological fluid with a general equation of state. Analytical solutions for the cosmic scale factor are obtained for arbitrary values of ω. Special cases including radiation, matter, stiff matter, and vacuum energy are analyzed. The results show that although the spacetime metric maintains the Friedmann–Lemaître–Robertson–Walker (FLRW) form, the expansion behavior of the universe depends strongly on the value of the equation-of-state

parameter. These results provide a unified theoretical description of the major stages of cosmic evolution.

Abstract

The large-scale structure and dynamics of the universe can be modeled by a relativistic perfect fluid characterized by an energy density E_d and pressure P_g. These quantities are related through the equation of state $P_g = \omega E_d$, where ω is the equation-of-state parameter. In this work we investigate the influence of the parameter ω on the spacetime geometry and cosmic expansion dynamics within the framework of general relativity. Starting from Einstein's field equations and assuming a homogeneous and isotropic universe, we derive the Friedmann equations for a cosmic fluid with both constant and time-dependent equation-of-state parameters. Analytical

solutions for the scale factor are obtained for several cosmological regimes including radiation, matter, stiff matter, and vacuum energy. Furthermore, we extend the standard cosmological model by introducing a time-dependent equation-of-state parameter $\omega(t)$. The resulting framework provides a unified description of transitions between cosmological epochs and may offer insights into the dynamics of dark energy.

1. Introduction

The modern understanding of cosmology is based on Einstein's theory of general relativity, which describes gravity as the curvature of spacetime caused by matter and energy. Observations of the cosmic microwave background radiation, large-scale galaxy distributions, and Type Ia supernovae strongly support the idea that the universe is homogeneous and isotropic on sufficiently large scales.
This assumption, known as the cosmological principle, allows the large-scale structure of the universe to be described by a simple spacetime geometry and a small number of physical parameters.

In relativistic cosmology the matter–energy content of the universe is modeled as a perfect fluid characterized by its energy density and pressure. The relationship between these quantities is described by an equation of state parameter **ω**, which determines the physical behavior of the cosmic fluid.

Understanding how the equation-of-state parameter **ω** affects cosmic expansion is essential for explaining the different phases of the universe, including:

- the radiation-dominated era
- the matter-dominated era
- the dark-energy-dominated era.

In this work we analyze the cosmological dynamics associated with a generalized equation of state and investigate how different values of the parameter **ω** influence spacetime evolution.

2. Energy–Momentum Tensor of a Perfect Fluid

The distribution of matter and energy in spacetime is described by the energy–momentum tensor. For a relativistic perfect fluid the tensor is given by

$$T_{\mu\nu} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$$

where

- E_d is the energy density
- P_g is the pressure
- u_μ is the four–velocity of the fluid
- $g_{\mu\nu}$ is the spacetime metric tensor.

The equation of state relating pressure and energy density is

$$P_g = \omega E_d$$

where ω is a dimensionless parameter that characterizes the physical nature of the cosmic fluid.

3. Einstein Field Equations

The gravitational dynamics of spacetime are governed by Einstein's field equations

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

where

- $G_{\mu\nu}$ is the Einstein curvature tensor
- G is Newton's gravitational constant
- c is the speed of light.

Including the cosmological constant Λ, the field equations become

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

These equations determine how the energy content of the universe affects spacetime geometry.

4. The FLRW Metric

Under the cosmological principle, the spacetime geometry of the universe is described by the Friedmann–Lemaître–Robertson–Walker metric

$$ds^2 = -c^2dt^2 + a(t)^2 \left[\frac{dr^2}{1-kr^2} + r^2(d\theta^2 + \sin^2\theta d\phi^2) \right]$$

where

- $a(t)$ is the cosmic scale factor
- k represents spatial curvature.

The values of k correspond to different geometries:

$$k = 0 \quad \text{flat universe}$$

$$k = +1 \quad \text{closed universe}$$

$$k = -1 \quad \text{open universe}$$

Research Note: Total Mass-Equivalent Density and Total Energy Density in Cosmological Expansion

1. Fundamental Relationship

In my cosmological framework, the total mass-equivalent density and total energy density are related through Einstein's mass–energy equivalence principle:

$E_d = \rho c^2$

Therefore:

$\rho = E_d/c^2$

Here, ρ represents the total mass-equivalent density, while E_d represents the associated total energy density. This relationship provides a direct connection between the density

representation of the cosmic energy content and the energy-density representation used in the Friedmann equations.

2. First Friedmann Equation

When the cosmological constant Λ is retained explicitly and its vacuum-energy contribution is excluded from E_d, the first Friedmann equation in SI units is:

$$H^2 = (8\pi G E_d)/(3c^2) - kc^2/a^2 + \Lambda c^2/3$$

This equation describes the relationship between the expansion rate of the universe, its energy density, spatial curvature, and the cosmological constant.

3. Second Friedmann Equation

Under the same energy-accounting convention, the second Friedmann equation is:

$$\ddot{a}/a = -(4\pi G/3c^2)(E_d + 3P_g) + \Lambda c^2/3$$

Using my equation of state,

$$P_g = \omega E_d$$

the acceleration equation becomes:

$$\ddot{a}/a = -(4\pi G E_d/3c^2)(1 + 3\omega) + \Lambda c^2/3$$

This form explicitly connects cosmic acceleration with total energy density, gravitational fluid pressure, the equation-of-state parameter, and the cosmological constant.

4. Physical Significance

The relationship $E_d = \rho c^2$ allows the density term in the Friedmann equations to be expressed consistently in terms of energy density. The first Friedmann equation describes the expansion rate, while the second describes the acceleration or deceleration of the cosmic scale factor.

The parameter ω characterizes the pressure-to-energy-density ratio:

$$\omega = P_g/E_d$$

Its value influences the pressure contribution to the acceleration equation. The cosmological constant contributes separately when it is retained explicitly in the field equations.

5. Consistency of Energy Accounting

For these equations, E_d must be defined consistently as the total energy density of the fluid components excluding the vacuum-energy contribution already represented by the explicit Λ term.

If the cosmological-constant energy density is instead included within E_d, the equations must be reformulated to prevent double-counting.

The cosmological-constant energy density is:

$$E_\Lambda = \Lambda c^4/(8\pi G)$$

and its pressure satisfies:

$$P_\Lambda = -E_\Lambda$$

Conclusion

The relationship $E_d = \rho c^2$ establishes the connection between total mass-equivalent density and its associated total energy density. Together with the first and second Friedmann equations, it provides a unified mathematical description of how energy density, pressure, spatial curvature, and the cosmological constant enter the dynamics of cosmic expansion.

This note establishes the definitions and equations used in my framework. Their full physical interpretation depends on maintaining consistent component definitions and energy accounting throughout the cosmological model.

Taking natural unit c=1,

5. Friedmann Equations

Applying Einstein's equations to the FLRW metric leads to the **Friedmann equations**.

First Friedmann equation:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3} E_d - \frac{k}{a^2} + \frac{\Lambda}{3}$$

Second Friedmann equation:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(E_d + 3P_g) + \frac{\Lambda}{3}$$

Substituting the equation of state gives

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3} E_d(1 + 3\omega) + \frac{\Lambda}{3}$$

6. Energy Conservation

Energy–momentum conservation implies

$$\nabla_\mu T^{\mu\nu} = 0$$

This leads to the continuity equation

$$\dot{E}_d + 3\frac{\dot{a}}{a}(E_d + P_g) = 0$$

Substituting $P_g = \omega E_d$

$$\dot{E}_d + 3(1+\omega)\frac{\dot{a}}{a}E_d = 0$$

The solution of this equation is

$$E_d = E_0 a^{-3(1+\omega)}$$

7. Evolution of the Scale Factor

For a spatially flat universe ($k = 0$) the Friedmann equation becomes

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3} E_d$$

Substituting the density evolution gives

$$a(t) \propto t^{\frac{2}{3(1+\omega)}}$$

8. Cosmological Regimes

Different values of ω correspond to different physical states of matter.

Radiation dominated universe:

$$\omega = \frac{1}{3}$$

$$a(t) \propto t^{1/2}$$

Matter dominated universe:

$$\omega = 0$$

$$a(t) \propto t^{2/3}$$

Stiff matter:

$$\omega = 1$$

$$a(t) \propto t^{1/3}$$

Vacuum energy:

$$\omega = -1$$

$$a(t) \propto e^{Ht}$$

---

9. Variable Equation of State

To generalize the model we introduce a time-dependent parameter

$$\omega = \omega(t)$$

The conservation equation becomes

$$\dot{E}_d + 3(1+\omega(t))\frac{\dot{a}}{a}E_d = 0$$

Integrating gives

$$E_d = E_0 \exp\left[-3\int(1+\omega(t))\frac{\dot{a}}{a}dt\right]$$

This formulation allows transitions between different cosmological phases.

10. Transition Between Cosmological Eras

A simple model for the evolving equation of state is

$$\omega(t) = \tfrac{1}{3} e^{-t/t_0}$$

This describes a transition from radiation domination toward matter domination and eventually dark-energy domination.

11. Dark Energy and Accelerated Expansion

From the second Friedmann equation

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3} E_d(1 + 3\omega)$$

Acceleration occurs when

$$\omega < -\frac{1}{3}$$

This condition corresponds to dark-energy-driven expansion.

12. Observational Implications

Observational measurements from supernova surveys and cosmic microwave background experiments suggest that the present universe is dominated by dark energy with

$$\omega \approx -1$$

This value corresponds to the cosmological constant model.

13. Discussion

The variable equation-of-state model provides a natural explanation for transitions between different cosmological epochs. Instead of treating radiation, matter, and dark energy as independent components, the cosmic fluid can evolve continuously through different thermodynamic states.

Such models may provide deeper insight into the origin of dark energy and the dynamical evolution of the universe.

14. Conclusion

In this work we investigated the spacetime dynamics of a cosmological fluid described by the equation of state $P_g = \omega E_d$. Using Einstein's field equations and the FLRW metric we derived the expansion laws for different cosmological regimes.

We extended the model by introducing a time-dependent equation-of-state parameter. The resulting framework provides a unified description of radiation, matter, and dark-energy dominated phases of cosmic evolution.

References

Einstein, A. (1916). *The Foundation of the General Theory of Relativity*.

Weinberg, S. *Cosmology*.

Carroll, S. *Spacetime and Geometry*.

Mukhanov, V. *Physical Foundations of Cosmology*.

Peebles, P. J. E. *Principles of Physical Cosmology*.

15. Reformulation of the Cosmological Constant for $\omega = -1$

In relativistic cosmology, the cosmological constant Λ can be interpreted as a form of vacuum energy. The equation-of-state parameter associated with vacuum energy is

$$\omega = -1$$

Using the equation of state

$$P_g = \omega E_d$$

the pressure of the vacuum becomes

$$P_g = -E_d$$

This relation implies that vacuum energy has **negative pressure equal in magnitude to its energy density**.

15.1 Vacuum Energy and the Cosmological Constant

Einstein's field equations with a cosmological constant are

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

The cosmological constant term can be rewritten as an effective energy–momentum tensor:

$$T^{(\Lambda)}_{\mu\nu} = -\frac{\Lambda c^4}{8\pi G} g_{\mu\nu}$$

Comparing this with the perfect fluid energy–momentum tensor

$$T_{\mu\nu} = (E_d + P_g) u_\mu u_\nu + P_g g_{\mu\nu}$$

we identify the vacuum energy density

$$E_\Lambda = \frac{\Lambda c^4}{8\pi G}$$

and the corresponding pressure

$$P_\Lambda = -E_\Lambda$$

Thus

$$\omega_\Lambda = \frac{P_\Lambda}{E_\Lambda} = -1$$

which confirms that the cosmological constant behaves as a perfect fluid with equation-of-state parameter $\omega = -1$.

15.2 Reformulated Cosmological Constant Using E_d

Using the energy density notation introduced in this work, the cosmological constant can be expressed directly in terms of E_d:

$$\Lambda = \frac{8\pi G}{c^4} E_d$$

for the case

$$\omega = -1$$

This expression establishes a direct connection between the cosmological constant and the vacuum energy density.

15.3 Modified Friedmann Equation

Substituting the vacuum energy density into the Friedmann equation yields

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3} E_d + \frac{\Lambda}{3} - \frac{k}{a^2}$$

Using the reformulated relation

$$\Lambda = \frac{8\pi G}{c^4} E_d$$

the equation becomes

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3} E_d + \frac{8\pi G}{3c^4} E_d - \frac{k}{a^2}$$

This formulation explicitly shows how the cosmological constant contributes to the cosmic expansion rate through the vacuum energy density.

NOTE: Dimensional and Unit Consistency of the Cosmic Expansion Equation and Cosmological Constant

1. Friedmann Expansion Equation

The cosmic expansion equation in SI units, without taking c = 1, is:

$(\dot{a}/a)^2 = (8\pi G\, E_d)/(3c^2) + (\Lambda c^2)/3 - kc^2/a^2$

Here:

$\dot{a}/a$ = Hubble expansion rate

G = Newton's gravitational constant

E_d = Energy density

Λ = Cosmological constant

k = Spatial-curvature parameter

a = Cosmic scale factor

c = Speed of light in vacuum

2. Proposed Relation for the Cosmological Constant

The cosmological constant is related to vacuum energy density by:

$\Lambda = (8\pi G/c^4)\, E_\Lambda$

Equivalently:

$E_\Lambda = \Lambda c^4/(8\pi G)$

Here, E_Λ is the vacuum energy density, which is distinct from the Energy density E_d.

3. Substitution into the Expansion Equation

Substituting the relation for Λ:

$(\dot{a}/a)^2 = (8\pi G\, E_d)/(3c^2) + (8\pi G\, E_\Lambda)/(3c^2) - kc^2/a^2$

Therefore:

$(\dot{a}/a)^2 = (8\pi G/3c^2)(E_d + E_\Lambda) - kc^2/a^2$

4. Units and Dimensions

Hubble expansion rate:

Unit = s^{-1}

Dimension = T^{-1}

Energy densities E_d and E_Λ:

Unit = J/m^3

Dimension = $ML^{-1}T^{-2}$

Newton's gravitational constant G:

Unit = $m^3/(kg \cdot s^2)$

Dimension = $L^3M^{-1}T^{-2}$

Speed of light c:

Unit = m/s

Dimension = LT^{-1}

Cosmological constant Λ:

Unit = m^{-2}

Dimension = L^{-2}

5. Dimensional Consistency

For the energy-density term:

$[(G\, E_d)/c^2] = T^{-2}$

For the cosmological-constant term:

$[\Lambda c^2] = T^{-2}$

For the curvature term, using a with length units and dimensionless k:

$[kc^2/a^2] = T^{-2}$

The left-hand side also has dimension:

$[(\dot{a}/a)^2] = T^{-2}$

Therefore, all terms in the Friedmann expansion equation have matching dimensions.

The proposed relation for the cosmological constant is also dimensionally consistent:

$[(G E_\Lambda)/c^4] = L^{-2} = [\Lambda]$

6. Conclusion

The dimensionally consistent expansion equation is:

$(\dot{a}/a)^2 = (8\pi G/3c^2)(E_d + E_\Lambda) - kc^2/a^2$

The cosmological constant is related to vacuum energy density by:

$\Lambda = (8\pi G/c^4) E_\Lambda$

The energy density E_d and vacuum energy density E_Λ must be distinguished to avoid counting the same energy contribution twice.

RESEARCH NOTE: UNIT AND DIMENSIONAL CONSISTENCY OF THE SECOND FRIEDMANN EQUATION

1. **SECOND FRIEDMANN EQUATION**

The cosmic acceleration equation, including the cosmological constant,without taking c = 1, is:

$(\ddot{a}/a) = -(4\pi G/3c^2)(E_d + 3P_g) + \Lambda c^2/3$

Where:

a = Cosmic scale factor

G = Newton's gravitational constant

E_d = Energy density

P_g = Isotropic gravitational pressure

Λ = Cosmological constant

c = Speed of light in vacuum

2. MY PROPOSED GRAVITATIONAL PRESSURE RELATIONSHIP

My proposed relationship is:

$P_g = \omega E_d$

Substituting into the second Friedmann equation:

$(\ddot{a}/a) = -(4\pi G/3c^2)(E_d + 3\omega E_d) + \Lambda c^2/3$

Taking E_d as a common factor:

$(\ddot{a}/a) = -(4\pi G E_d/3c^2)(1 + 3\omega) + \Lambda c^2/3$

3. COSMOLOGICAL CONSTANT AND VACUUM ENERGY DENSITY

The cosmological constant is related to its equivalent vacuum energy density by:

$\Lambda = (8\pi G/c^4)E_\Lambda$

Therefore:

$E_\Lambda = \Lambda c^4/(8\pi G)$

Substituting this relationship into the acceleration equation:

$(\ddot{a}/a) = -(4\pi G/3c^2)(E_d + 3P_g) + (c^2/3)(8\pi G E_\Lambda/c^4)$

Simplifying:

$(\ddot{a}/a) = -(4\pi G/3c^2)(E_d + 3P_g) + 8\pi G E_\Lambda/(3c^2)$

Therefore:

$(\ddot{a}/a) = -(4\pi G/3c^2)(E_d + 3P_g - 2E_\Lambda)$

Using my proposed relationship $P_g = \omega E_d$:

$(\ddot{a}/a) = -(4\pi G/3c^2)[E_d(1 + 3\omega) - 2E_\Lambda]$

4. UNITS AND DIMENSIONS

Cosmic acceleration per unit scale factor:

Unit: s^{-2}

Dimension: T^{-2}

Newton's gravitational constant:

Unit: $m^3/(kg \cdot s^2)$

Dimension: $L^3M^{-1}T^{-2}$

Energy density and pressure:

Unit: J/m^3

Dimension: $ML^{-1}T^{-2}$

Speed of light:

Unit: m/s

Dimension: LT^{-1}

Cosmological constant:

Unit: m^{-2}

Dimension: L^{-2}

Proportionality parameter:

ω is dimensionless.

5. DIMENSIONAL CONSISTENCY

For the energy-density and pressure term:

$(GE_d)/c^2$ has dimension T^{-2}.

For the cosmological-constant term:

Λc^2 has dimension T^{-2}.

For the vacuum energy-density term:

$(GE_\Lambda)/c^2$ has dimension T^{-2}.

Therefore, every term in the acceleration equation has dimension T^{-2}, matching the left-hand side.

6. CONCLUSION

The second Friedmann equation is:

$(\ddot{a}/a) = -(4\pi G/3c^2)(E_d + 3P_g) + \Lambda c^2/3$

Using my proposed pressure relationship:

$P_g = \omega E_d$

The equation becomes:

$(\ddot{a}/a) = -(4\pi G E_d/3c^2)(1 + 3\omega) + \Lambda c^2/3$

Expressing the cosmological constant through its equivalent vacuum energy density gives:

$(\ddot{a}/a) = -(4\pi G/3c^2)(E_d + 3P_g - 2E_\Lambda)$

These equations are dimensionally consistent in SI units, provided energy densities and pressures are defined consistently and the vacuum energy contribution is not counted twice.

This analysis verifies the algebra and dimensional consistency of the equations. It does not independently establish whether my proposed gravitational pressure represents a new physical source in cosmology.

Special Case: Cosmological Constant for ω = −1

In this research, the equation of state of the cosmological fluid is defined as:

$P_g = \omega E_d$

For the special case $\omega = -1$, the equation becomes:

$P_g = -E_d$

When the cosmological fluid represents only the cosmological-constant component, its total energy density is equal to the cosmological-constant energy density:

$E_d = E_\Lambda$

In SI units, the cosmological-constant energy density is defined as:

$E_\Lambda = \Lambda c^4/(8\pi G)$

Therefore:

$E_d = \Lambda c^4/(8\pi G)$

The corresponding pressure is:

$P_g = -E_d = -\Lambda c^4/(8\pi G)$

Here, E_d is the total energy density of the fluid identified with the cosmological constant, E_Λ is the cosmological-constant energy density, P_g is the fluid pressure, Λ is the cosmological constant, c is the speed of light, and G is Newton's gravitational constant.

Thus, when $\omega = -1$ and the fluid is identified with the cosmological constant, its pressure is equal in magnitude and opposite in sign to its energy density. The energy density and pressure are expressed in joules per cubic metre (J/m^3).

Separate Note: Component Identification for the Cosmological Constant When ω = −1

The equation of state of the cosmological fluid is defined as:

$P_g = \omega E_d$

For the special case $\omega = -1$:

$P_g = -E_d$

When the fluid represented by E_d is identified specifically as the cosmological-constant component, its energy density is:

$E_d = E_\Lambda$

In SI units:

$E_d = E_\Lambda = \Lambda c^4/(8\pi G)$

Consequently, the pressure becomes:

$P_g = -E_d = -\Lambda c^4/(8\pi G)$

Here, E_d represents the energy density of the identified fluid component, E_Λ represents the cosmological-constant energy density, P_g is the corresponding pressure, Λ is the cosmological constant, c is the speed of light, and G is Newton's gravitational constant.

The component-identification condition distinguishes cosmological-constant energy from a combined energy density containing multiple components, such as matter, radiation, and vacuum energy. If E_d represents a combined fluid, the equality $E_d = E_\Lambda$ does not follow from $\omega = -1$ alone. It requires the fluid to be identified with the cosmological-constant component.

Thus, for the cosmological-constant component, the equation-of-state parameter is $\omega = -1$, and its pressure is equal in magnitude and opposite in sign to its energy density.

16. Comparison with Standard Cosmological Constant Model

The standard cosmological model treats the cosmological constant as an independent parameter. In contrast, the present formulation expresses the cosmological constant directly in terms of the energy density associated with the equation-of-state parameter $\omega = -1$.

Standard Model

$$\Lambda = \text{constant}$$

Vacuum energy density:

$$E_\Lambda = \frac{\Lambda c^4}{8\pi G}$$

Reformulated Model

$$\Lambda = \frac{8\pi G}{c^4} E_d$$

with

$$\omega = -1$$

Key Differences

Aspect
Cosmological constant
Vacuum equation of state
Physical interpretation

Standard Cosmology
independent parameter
$\omega = -1$ assumed
constant curvature term

Reformulated Model
derived from energy density
$\omega = -1$ fundamental condition
vacuum energy density of cosmic fluid

17. Physical Implications

The reformulation suggests that the cosmological constant may not be an independent geometric parameter but rather a manifestation of the vacuum energy density of the cosmic fluid.

This interpretation provides a natural explanation for dark energy as a special thermodynamic state characterized by

$$\omega = -1$$

Such a framework may help bridge the gap between cosmology and quantum vacuum physics.

18. Conclusion of the Extended Model

By reformulating the cosmological constant in terms of the equation-of-state parameter $\omega = -1$, we obtain a direct connection between vacuum energy and the cosmological constant. This approach emphasizes the role of energy density in determining the large-scale dynamics of the universe and provides an alternative perspective on the origin of dark energy.

19. Derivation of a Generalized Friedmann Equation Using $\omega(t)$ and E_d

In standard cosmology the equation-of-state parameter ω is usually treated as a constant. However, observational evidence related to dark energy suggests that the equation-of-state parameter may vary with cosmic time.

To account for this possibility we introduce a **time-dependent equation-of-state parameter**

$$\omega = \omega(t)$$

The pressure of the cosmic fluid becomes

$$P_g = \omega(t) E_d$$

19.1 Continuity Equation with Variable Equation of State

Energy–momentum conservation requires

$$\nabla_\mu T^{\mu\nu} = 0$$

For the FLRW metric this leads to the continuity equation

$$\dot{E}_d + 3\frac{\dot{a}}{a}(E_d + P_g) = 0$$

Substituting the equation of state

$$P_g = \omega(t) E_d$$

gives

$$\dot{E}_d + 3(1 + \omega(t))\frac{\dot{a}}{a} E_d = 0$$

19.2 General Energy Density Evolution

Rearranging the continuity equation,

$$\frac{\dot{E}_d}{E_d} = -3(1+\omega(t))\frac{\dot{a}}{a}$$

Integrating over time,

$$E_d = E_0 \exp\left[-3\int(1+\omega(t))\frac{\dot{a}}{a}dt\right]$$

This expression represents the **general evolution of cosmic energy density for a variable equation-of-state parameter**.

19.3 Generalized Friedmann Equation

The standard Friedmann equation is

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3} E_d - \frac{k}{a^2} + \frac{\Lambda}{3}$$

Substituting the generalized energy density expression,

$$E_d = E_0 \exp\left[-3 \int (1 + \omega(t)) \frac{\dot{a}}{a} dt\right]$$

we obtain the **generalized Friedmann equation**

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3} E_0 \exp\left[-3 \int (1 + \omega(t)) \frac{\dot{a}}{a} dt\right] - \frac{k}{a^2} + \frac{\Lambda}{3}$$

19.4 Special Case: Constant ω

If $\omega(t) = \omega$ is constant, the integral simplifies and the standard result is recovered

$$E_d = E_0 a^{-3(1+\omega)}$$

Substituting into the Friedmann equation gives

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3} E_0 a^{-3(1+\omega)} - \frac{k}{a^2} + \frac{\Lambda}{3}$$

Thus the generalized equation correctly reduces to the classical Friedmann cosmology.

20. Physical Interpretation of the Generalized Model

The generalized Friedmann equation shows that the expansion of the universe depends not only on the energy density but also on the **time evolution of the equation-of-state parameter**.

Several cosmological scenarios can be described using this framework:

1. **Early universe**

$\omega = \frac{1}{3}$

radiation dominated expansion.

2. **Matter dominated epoch**

$$\omega = 0$$

gravitational structure formation.

3. **Dark energy dominated era**

$$\omega \approx -1$$

accelerated cosmic expansion.

4. **Dynamical dark energy**

$$\omega = \omega(t)$$

slow evolution of vacuum energy.

21. Significance of the Generalized Friedmann Equation

The generalized equation provides several advantages:

• describes transitions between cosmological eras
• allows modeling of dynamical dark energy
• unifies different cosmological regimes in a single framework

Such models are widely studied in modern cosmology to explain the observed accelerated expansion of the universe.

22. Final Remarks

By introducing a variable equation-of-state parameter $\omega(t)$ and deriving the corresponding generalized Friedmann equation, we obtain a more flexible description of cosmic evolution. This approach allows a unified treatment of radiation, matter, and dark energy within a single theoretical framework.

23. Comparison with Observational Parameters (Planck Cosmology)

Modern cosmology is strongly constrained by high-precision observations of the cosmic microwave background (CMB). The most accurate measurements currently come from the Planck satellite, which has provided precise estimates of the fundamental cosmological parameters.

The theoretical framework developed in this work can be compared with the observational parameters determined by the Planck mission.

23.1 Hubble Constant

The Hubble parameter describes the expansion rate of the universe and is defined as

$$H = \frac{\dot{a}}{a}$$

The Planck observations give

$$H_0 \approx 67.4\ \mathrm{km\ s^{-1} Mpc^{-1}}$$

In the generalized Friedmann equation derived in this work,

$$H^2 = \frac{8\pi G}{3} E_d - \frac{k}{a^2} + \frac{\Lambda}{3}$$

the Hubble parameter depends directly on the energy density E_d and the cosmological constant.

This confirms that the expansion rate predicted by the model is consistent with observational cosmology.

23.2 Matter Density Parameter

The matter density parameter is defined as

$$\Omega_m = \frac{8\pi G E_m}{3H^2}$$

According to Planck observations

$$\Omega_m \approx 0.315$$

This value corresponds to the **matter-dominated era described by**

$$\omega = 0$$

which matches the matter-dominated cosmological regime in the model.

23.3 Dark Energy Density Parameter

The dark energy density parameter is

$$\Omega_\Lambda = \frac{\Lambda}{3H^2}$$

Planck measurements give

$$\Omega_\Lambda \approx 0.685$$

In the framework developed in this work, dark energy corresponds to the case

$$\omega = -1$$

with

$$P_g = -E_d$$

This confirms that the cosmological constant behaves as a vacuum energy component.

23.4 Radiation Density Parameter

Radiation density is defined as

$$\Omega_r = \frac{8\pi G E_r}{3H^2}$$

Observations show

$$\Omega_r \approx 9 \times 10^{-5}$$

This corresponds to the radiation-dominated equation-of-state parameter

$$\omega = \frac{1}{3}$$

which is consistent with the early universe predictions of the model.

23.5 Spatial Curvature

The curvature parameter is

$$\Omega_k = -\frac{k}{a^2 H^2}$$

Planck observations indicate

$$\Omega_k \approx 0$$

which implies that the universe is spatially flat.

This is consistent with the flat-universe assumption used in many of the derivations in this work.

23.6 Dark Energy Equation of State

Observational studies constrain the dark-energy equation-of-state parameter.

Planck data suggests

$$\omega \approx -1$$

with only small deviations allowed.

This result supports the theoretical interpretation of the cosmological constant as vacuum energy with

$$P_g = -E_d$$

23.7 Consistency with the ΛCDM Model

The current standard cosmological model is the **ΛCDM model**, which includes:

- cold dark matter
- cosmological constant
- radiation
- baryonic matter.

The parameters obtained by the Planck mission are consistent with the cosmological framework developed in this work.

The equation-of-state parameter approach provides a unified way to describe these components:

Cosmic Component	Equation of State
Radiation	$\omega = 1/3$
Matter	$\omega = 0$
Stiff matter	$\omega = 1$
Dark energy	$\omega = -1$

23.8 Implications for the Generalized Model

The generalized model with a time-dependent equation-of-state parameter

$$\omega = \omega(t)$$

can naturally explain transitions between the different cosmological regimes observed in the universe.

This feature provides a possible theoretical framework for describing the evolution of dark energy and the long-term expansion of the universe.

24. Summary of Observational Comparison

The theoretical framework developed in this work is consistent with key cosmological observations obtained by the Planck mission:

Parameter
Hubble constant
Matter density
Dark energy density
Radiation density
Spatial curvature

Planck Observation
$H_0 \approx 67.4$ km/s/Mpc
$\Omega_m \approx 0.315$
$\Omega_\Lambda \approx 0.685$
$\Omega_r \approx 9 \times 10^{-5}$
$\Omega_k \approx 0$

Model Prediction
Derived from Friedmann equation
$\omega = 0$ regime
$\omega = -1$ vacuum energy
$\omega = 1/3$
Flat universe assumption

The agreement between theoretical predictions and observational measurements supports the validity of the cosmological fluid approach used in this study.

Derivation of the Cosmological Constant from the Equation-of-State Condition $\omega = -1$

In the cosmological fluid description used in this work, the pressure and energy density are related through

$$P_g = \omega E_d$$

For vacuum energy the equation-of-state parameter satisfies

$$\omega = -1$$

Thus the pressure becomes

$$P_g = -E_d$$

This relation characterizes the thermodynamic behavior of vacuum energy.

Energy–Momentum Tensor Comparison

The energy–momentum tensor of a perfect fluid is

$$T_{\mu\nu} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$$

Substituting $P_g = -E_d$:

$$T_{\mu\nu} = (E_d - E_d)u_\mu u_\nu - E_d g_{\mu\nu}$$

which simplifies to

$$T_{\mu\nu} = -E_d g_{\mu\nu}$$

Einstein Field Equation

Einstein's equation with the cosmological constant is

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

The cosmological constant term can be moved to the right-hand side:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} \left(T_{\mu\nu} + T_{\mu\nu}^{(\Lambda)} \right)$$

where

$$T_{\mu\nu}^{(\Lambda)} = -\frac{\Lambda c^4}{8\pi G} g_{\mu\nu}$$

Identification of Vacuum Energy Density

Comparing

$$T_{\mu\nu} = -E_d g_{\mu\nu}$$

with

$$T_{\mu\nu}^{(\Lambda)} = -\frac{\Lambda c^4}{8\pi G} g_{\mu\nu}$$

we obtain

$$E_d = \frac{\Lambda c^4}{8\pi G}$$

or equivalently

$$\Lambda = \frac{8\pi G}{c^4} E_d$$

Interpretation

This result shows that the cosmological constant can be interpreted as the **vacuum energy density of the cosmic fluid** corresponding to the equation-of-state parameter

$$\omega = -1$$

Thus the cosmological constant emerges naturally as a special thermodynamic state of the cosmic fluid.

Significance for the Cosmological Model

The derivation demonstrates that:

1. The cosmological constant corresponds to vacuum energy.
2. Vacuum energy obeys the equation of state $P = -\rho$.
3. The cosmological constant can be expressed directly in terms of energy density:

$$\Lambda = \frac{8\pi G}{c^4} E_d$$

This relation provides a direct connection between spacetime geometry and the thermodynamic properties of the cosmic fluid.

Cosmological Constant and Cosmic Acceleration

In the cosmological fluid description used in this work, the pressure and energy density satisfy the equation of state

$$P_g = \omega E_d$$

For the vacuum energy case

$$\omega = -1$$

so the pressure becomes

$$P_g = -E_d$$

This negative pressure is responsible for the accelerated expansion of the universe.

Acceleration Equation

The cosmic acceleration is determined by the **second Friedmann equation**

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(E_d + 3P_g)$$

Substituting

$$P_g = \omega E_d$$

we obtain

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}E_d(1 + 3\omega)$$

Condition for Accelerated Expansion

Acceleration occurs when

$$\frac{\ddot{a}}{a} > 0$$

which requires

$$1 + 3\omega < 0$$

Thus

$$\omega < -\frac{1}{3}$$

Special Case: Cosmological Constant

For vacuum energy

$$\omega = -1$$

so

$$1 + 3(-1) = -2$$

which gives

$$\frac{\ddot{a}}{a} > 0$$

Therefore the universe undergoes **accelerated expansion**.

Relation to the Cosmological Constant

From your derivation

$$\Lambda = \frac{8\pi G}{c^4} E_d$$

the Friedmann equation becomes

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3} E_d + \frac{\Lambda}{3} - \frac{k}{a^2}$$

Since E_d is constant when $\omega = -1$, the solution for the scale factor is

$$a(t) \propto e^{Ht}$$

which corresponds to **exponential cosmic expansion** (de Sitter universe).

Physical Meaning

Your derivation shows that

- $\omega = -1$ corresponds to **vacuum energy**
- vacuum energy produces **negative pressure**
- negative pressure drives **cosmic acceleration**
- the cosmological constant can therefore be interpreted as **vacuum energy density**

This is the physical mechanism behind the **observed accelerated expansion of the universe**.

The cosmological constant corresponds to a cosmic fluid with equation-of-state parameter $\omega = -1$, which naturally leads to accelerated expansion of the universe.

Phase Evolution of the Universe: $\omega = \frac{1}{3} \to 0 \to -1$

The large-scale evolution of the universe can be described through different phases characterized by the equation-of-state parameter ω of the cosmic fluid. The relation between pressure and energy density is

$$P_g = \omega E_d$$

Different values of ω correspond to different dominant components of the universe. Observations indicate that the universe evolves through three major cosmological phases: radiation domination, matter domination, and dark-energy domination.

1. Radiation-Dominated Phase

In the early universe, radiation was the dominant component. For radiation the equation-of-state parameter is

$$\omega = \frac{1}{3}$$

Thus

$$P_g = \frac{1}{3} E_d$$

The scale factor evolves as

$$a(t) \propto t^{1/2}$$

During this phase:

- photons and relativistic particles dominate
- the universe expands rapidly
- temperature decreases with expansion

This phase occurred roughly during the first **50,000 years** after the Big Bang.

2. Matter-Dominated Phase

As the universe expanded and cooled, matter began to dominate the cosmic energy density. In this regime

$$\omega = 0$$

Thus

$$P_g = 0$$

Matter behaves as pressureless dust. The scale factor evolves as

$$a(t) \propto t^{2/3}$$

This phase allowed gravitational instability to form structures such as

- galaxies
- galaxy clusters
- large-scale cosmic structures.

The matter-dominated era lasted for billions of years.

3. Dark-Energy-Dominated Phase

In the present universe, observations indicate that dark energy dominates the total energy density. Dark energy corresponds to

$$\omega = -1$$

so that

$$P_g = -E_d$$

This negative pressure drives cosmic acceleration.

The scale factor evolves exponentially:

$$a(t) \propto e^{Ht}$$

This corresponds to the **de Sitter expansion** of the universe.

4. Evolution of the Equation-of-State Parameter

The cosmic evolution can therefore be summarized as a transition of the equation-of-state parameter

$$\omega = \frac{1}{3} \rightarrow 0 \rightarrow -1$$

This sequence describes the transition from

- radiation domination
- matter domination
- dark energy domination.

5. Unified Cosmological Interpretation

In the cosmological fluid model used in this work, the evolution of the universe can be interpreted as a gradual change in the thermodynamic state of the cosmic fluid.

The equation-of-state parameter acts as a fundamental quantity describing the physical behavior of the dominant cosmic component.

Thus the expansion history of the universe is determined by the evolution of ω.

6. Cosmological Significance

This phase evolution framework provides a unified description of cosmic history and explains several observational features of the universe:

- early universe dominated by radiation
- structure formation during the matter era
- present accelerated expansion due to dark energy.

Summary

The evolution of the universe can be represented by the sequence

$$\omega = \frac{1}{3} \rightarrow 0 \rightarrow -1$$

which corresponds to the transition

$$\text{Radiation} \rightarrow \text{Matter} \rightarrow \text{Dark Energy}$$

This framework provides a consistent description of the major phases of cosmic evolution within the relativistic cosmological fluid model.

Future Evolution of the Universe for $\omega \leq -1$

The late-time behavior of the universe depends strongly on the value of the equation-of-state parameter ω of the dominant cosmic component. Observational evidence indicates that the present universe is dominated by dark energy with an equation-of-state parameter close to

$$\omega \approx -1$$

However, theoretical models allow the possibility that

$$\omega < -1$$

which corresponds to a hypothetical form of dark energy known as **phantom energy**.

1. Accelerated Expansion

The acceleration of the universe is governed by the second Friedmann equation

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(E_d + 3P_g)$$

Substituting the equation of state

$$P_g = \omega E_d$$

we obtain

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}E_d(1 + 3\omega)$$

Acceleration occurs when

$$\omega < -\frac{1}{3}$$

This condition is satisfied for dark energy.

2. Cosmological Constant Case $(\omega = -1)$

For the cosmological constant,

$$\omega = -1$$

the energy density remains constant

$$E_d = \text{constant}$$

The scale factor evolves exponentially

$$a(t) \propto e^{Ht}$$

This corresponds to the **de Sitter universe**, where the expansion rate remains constant.

In this scenario

- galaxies gradually move beyond the observable horizon
- the universe becomes increasingly empty
- cosmic expansion continues indefinitely.

3. Phantom Energy $(\omega < -1)$

If the equation-of-state parameter satisfies

$$\omega < -1$$

the energy density increases as the universe expands.

From the conservation equation

$$E_d \propto a^{-3(1+\omega)}$$

when

$$\omega < -1$$

the exponent becomes negative, implying

$$E_d \propto a^{|3(1+\omega)|}$$

Thus the energy density grows with time.

This leads to a rapidly accelerating universe.

4. Big Rip Scenario

In phantom-energy cosmology, the scale factor diverges in a finite time. The universe eventually reaches a singular state known as the **Big Rip**.

As the expansion accelerates:

1. galaxy clusters become gravitationally unbound
2. galaxies are torn apart
3. solar systems are disrupted
4. atoms themselves may be destroyed.

Mathematically, the scale factor behaves approximately as

$$a(t) \propto (t_{\text{rip}} - t)^{-n}$$

where t_{rip} is the time of the Big Rip singularity.

5. Observational Constraints

Current observational data suggest that the equation-of-state parameter is close to

$$\omega \approx -1$$

with small uncertainties. This implies that the universe is most consistent with a cosmological constant model.

However, small deviations from $\omega = -1$ remain possible, leaving open the possibility of dynamical dark energy.

6. Implications for Cosmology

The future fate of the universe therefore depends on the value of the equation-of-state parameter:

Equation of State	Future of the Universe
$\omega > -1/3$	decelerating expansion
$-1 < \omega < -1/3$	accelerating expansion
$\omega = -1$	de Sitter expansion
$\omega < -1$	Big Rip scenario

Thus the equation-of-state parameter plays a central role in determining the ultimate fate of the universe.

Final Perspective

The cosmological fluid model developed in this work provides a unified framework for describing the entire history of the universe:

$$\omega = \frac{1}{3} \rightarrow 0 \rightarrow -1$$

corresponding to

$$\text{Radiation} \rightarrow \text{Matter} \rightarrow \text{Dark Energy}$$

and potentially extending to

$$\omega < -1$$

in the case of phantom-energy cosmology.

Consistency with Observations of Cosmic Acceleration

The theoretical framework developed in this work is consistent with modern cosmological observations indicating that the expansion of the universe is accelerating. Observational evidence for cosmic acceleration was first obtained from measurements of distant Type Ia supernovae and has since been confirmed by multiple independent observations including cosmic microwave background measurements and large-scale structure surveys.

In the cosmological fluid description used in this study, the expansion dynamics are governed by the second Friedmann equation

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(E_d + 3P_g)$$

Using the equation of state

$$P_g = \omega E_d$$

the acceleration equation becomes

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}E_d(1 + 3\omega)$$

From this expression it follows that accelerated expansion occurs when

$$\omega < -\frac{1}{3}$$

Modern observational data indicate that the present universe is dominated by dark energy with an equation-of-state parameter close to

$$\omega \approx -1$$

which corresponds to the cosmological constant model.

In the present framework, the case

$$\omega = -1$$

implies

$$P_g = -E_d$$

which produces negative pressure and leads naturally to accelerated cosmic expansion. The scale factor in this regime evolves approximately as

$$a(t) \propto e^{Ht}$$

representing the de Sitter expansion of the universe.

Therefore, the cosmological fluid model developed in this work, together with the reformulated cosmological constant relation

$$\Lambda = \frac{8\pi G}{c^4} E_d$$

provides a theoretical description that is consistent with current observational evidence for cosmic acceleration.

Summary and Outlook

In this work we investigated the dynamics of a cosmological fluid characterized by the equation of state

$$P_g = \omega E_d$$

within the framework of general relativity. Starting from Einstein's field equations and assuming a homogeneous and isotropic universe described by the Friedmann–Lemaître–Robertson–Walker metric, we derived the Friedmann equations governing the expansion of the universe.

The analysis shows that the equation-of-state parameter ω plays a central role in determining the expansion behavior of the universe. Different values of ω correspond to different cosmological regimes:

$$\omega = \tfrac{1}{3} \quad \text{radiation-dominated era}$$

$$\omega = 0 \quad \text{matter-dominated era}$$

$$\omega = -1 \quad \text{dark-energy-dominated era}$$

These regimes describe the observed phase evolution of the universe from radiation domination to matter domination and finally to the present epoch of accelerated expansion.

A reformulation of the cosmological constant was presented using the vacuum equation-of-state condition

$$\omega = -1$$

which leads to the relation

$$\Lambda = \frac{8\pi G}{c^4} E_d$$

This result shows that the cosmological constant can be interpreted as the vacuum energy density of the cosmic fluid. In this regime the pressure satisfies

$$P_g = -E_d$$

which produces negative pressure and leads naturally to cosmic acceleration.

Furthermore, a generalized Friedmann equation was derived by allowing the equation-of-state parameter to evolve with cosmic time

$$\omega = \omega(t)$$

This generalization provides a unified framework for describing transitions between different cosmological epochs and allows the possibility of dynamical dark energy.

The theoretical predictions of the model were compared with modern cosmological observations, including measurements of the cosmic microwave background and large-scale structure. The results are consistent with current observational constraints indicating that the universe is spatially flat and dominated by dark energy with an equation-of-state parameter close to

$$\omega \approx -1$$

Outlook

Although the cosmological constant model provides a simple explanation for cosmic acceleration, the physical origin of dark energy remains one of the most important open questions in modern cosmology.

The framework developed in this work suggests several possible directions for future research:

- investigation of time-dependent equation-of-state models
- exploration of connections between vacuum energy and quantum field theory
- observational constraints on dynamical dark-energy models
- theoretical studies of the long-term fate of the universe.

Further progress in both observational cosmology and theoretical physics will be essential for understanding the fundamental nature of dark energy and the ultimate evolution of the universe.

Note: In this Research Paper, ChatGpt response used word ‘your’ for Author (me).

Observational Evidence and the Cosmological Acceleration Equation

Present astronomical observations provide substantial evidence that the expansion of the universe has been accelerating. The landmark evidence came from observations of distant Type Ia supernovae in 1998. These supernovae appeared dimmer, and therefore farther away, than expected in a universe whose expansion was slowing under gravity. Subsequent cosmological investigations have continued to examine this conclusion. Recent analyses support cosmic acceleration, although some researchers continue to debate the interpretation of particular datasets.

The acceleration of the cosmic scale factor is described by the second Friedmann equation. In SI units, with the cosmological constant retained explicitly, the equation is:

$(\ddot{a}/a) = -(4\pi G/3c^2)(E_d + 3P_g) + \Lambda c^2/3$

Using the equation of state:

$P_g = \omega E_d$

the acceleration equation becomes:

$(\ddot{a}/a) = -(4\pi G\, E_d/3c^2)(1 + 3\omega) + \Lambda c^2/3$

Here, a is the cosmic scale factor, ä is its second derivative with respect to cosmic time, E_d is the energy density of the fluid components excluding the vacuum contribution represented explicitly by Λ, P_g is the fluid pressure, and ω is the equation-of-state parameter.

Observational evidence for accelerated expansion motivates the investigation of the physical contributions appearing in this equation. In the standard cosmological interpretation, a positive cosmological constant contributes positively to ä/a. A fluid with sufficiently negative pressure can also produce acceleration.

For the special case ω = −1, the equation of state becomes:

$P_g = -E_d$

If the fluid represents the cosmological-constant component alone, then:

$E_d = E_\Lambda = \Lambda c^4/(8\pi G)$

This special case describes vacuum energy with negative pressure. However, observations of accelerated expansion do not, by themselves, establish that every possible cosmological model must use a constant ω or that the underlying physical nature of dark energy has been conclusively identified.

Conclusion

The observed acceleration of cosmic expansion provides an important empirical context for examining the Friedmann equations and the equation-of-state parameter ω. A complete test of a cosmological model requires comparing its quantitative predictions with supernova distances, baryon acoustic oscillations, cosmic microwave background measurements, and other relevant observations.

Research Note: Accelerated Expansion of the Universe Without an Explicit Cosmological Constant

Present astronomical observations indicate that the expansion of the universe is accelerating. This observational result can be investigated using the second Friedmann equation without including an explicit cosmological constant.

In SI units, with Λ = 0, the acceleration equation is:

$(\ddot{a}/a) = -(4\pi G/3c^2)(E_d + 3P_g)$

Using the equation of state:

$P_g = \omega E_d$

the acceleration equation becomes:

$(\ddot{a}/a) = -(4\pi G\, E_d/3c^2)(1 + 3\omega)$

Here, a is the cosmic scale factor, E_d is the energy density of the cosmological fluid, P_g is its pressure, G is Newton's gravitational constant, c is the speed of light, and ω is the equation-of-state parameter.

For positive E_d, accelerated expansion requires:

ä > 0

Therefore:

1 + 3ω < 0

and hence:

ω < −1/3

This result shows that, within the standard Friedmann framework without an explicit cosmological constant, a fluid with sufficiently negative pressure can produce accelerated expansion.

For the special case ω = −1:

P_g = −E_d

Substituting this value into the acceleration equation gives:

(ä/a) = (8πG/3c²)E_d

For positive E_d, the acceleration is positive.

Conclusion

The Friedmann acceleration equation permits accelerated expansion without an explicit cosmological constant if the cosmic fluid has sufficiently negative pressure. This mathematical result provides a framework for investigating whether a fluid described by P_g = ωE_d can account for the observed acceleration.

However, observational evidence for acceleration alone does not establish that this particular fluid model is the correct physical explanation. Its predictions must be tested against cosmological observations.